Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 3.0, **Version Date** 7/9/13

HV Tank Feedback Resistor Verification

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	45 mins	Not Applicable

2 Overview

This procedure verifies the kV accuracy, by actually taking scans with connecting a high voltage bleeder.

Last Revised:	22 January, 2013

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
High Voltage bleeder	1	-	-	-
Oscilloscope	1	-	-	-

3.2 Safety



NOTICE

HV Cable Damage.

Do not sharply bend HV cables, or damage the cables. The minimum allowed radius of bent HV cables is 6.35 cm (2.5 in.).

4 Procedure



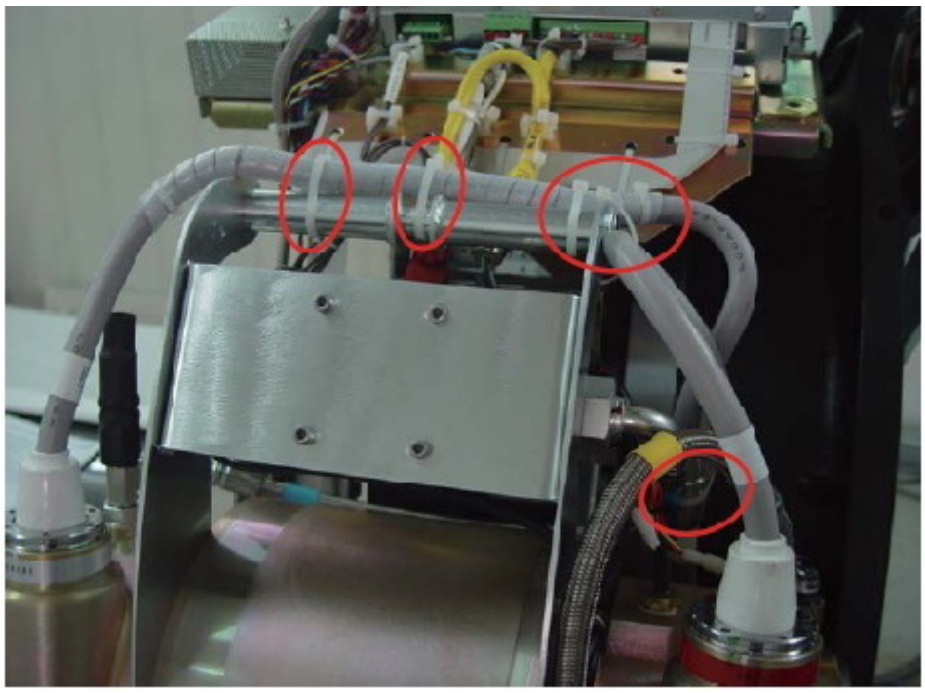
⚠ CAUTION

Electric shock

Use a ground bar to discharge the high voltage connector any time you disconnect it.

1. Perform warm-up scans, if the x-ray tube is cold.
2. Remove the gantry left side cover.
3. Turn off all three service switches on the Service Switch Panel (Axial Drive, HVDC, 120VAC)
4. Remove the right side, top and front gantry covers.
5. Rotate the gantry such that the x-ray tube is in the 3 o'clock position.
6. Lock the gantry in position, using the rotational lock. Ensure that gantry rotation is locked by attempting to rotate the gantry by hand.
7. Free the HV cables.
    a. Carefully cut tie-wraps securing HV cables. Note HV cable routing.

Illustration 1: Cut Tie-wraps by Securing HV Cables



b. Loosen the cable's locking rings with a spanner wrench, and pull HV cable terminal sticks (Anode/Cathode) out of the Tube.

c. Wipe excess oil from the HV cable terminal sticks.

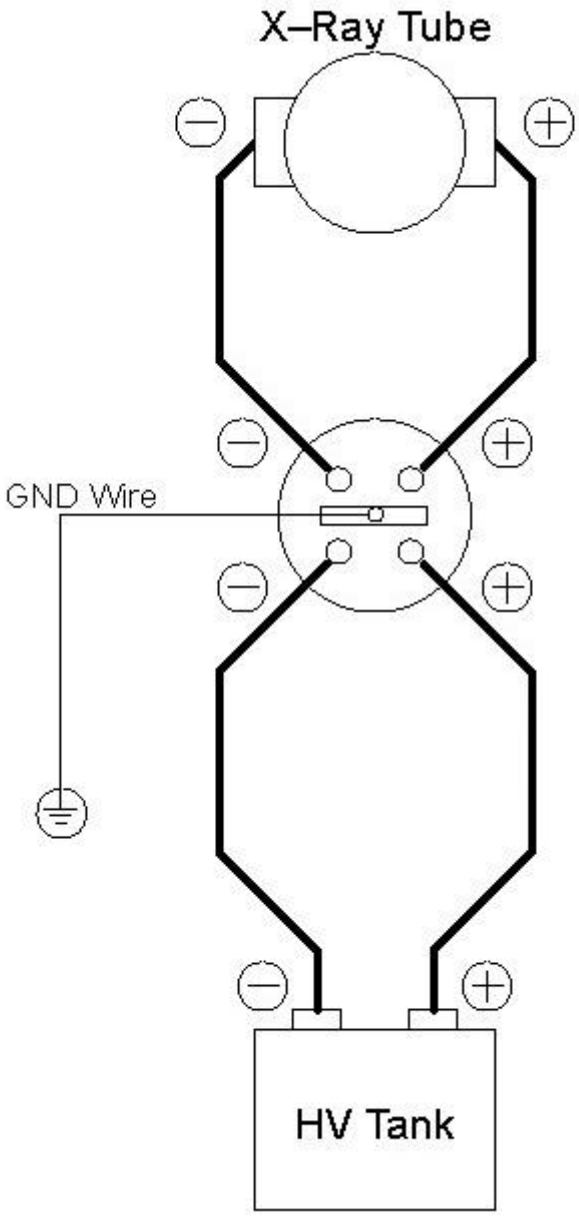
8. Install the HV bleeder between the HV tank and tube. Use the patient table to place the bleeder.

   a. Insert the HV cable terminal sticks (tube) into the receptacles of HV bleeder, and insert the HV cable terminal sticks (HV bleeder) into the receptacles of tube.

   b. Replenish the HV cable terminal sticks well of the receptacles with silicone oil.

   c. Tighten the HV cables with the spanner wrench and wipe excess oil

Illustration 2: Bleeder Connection Block Diagram



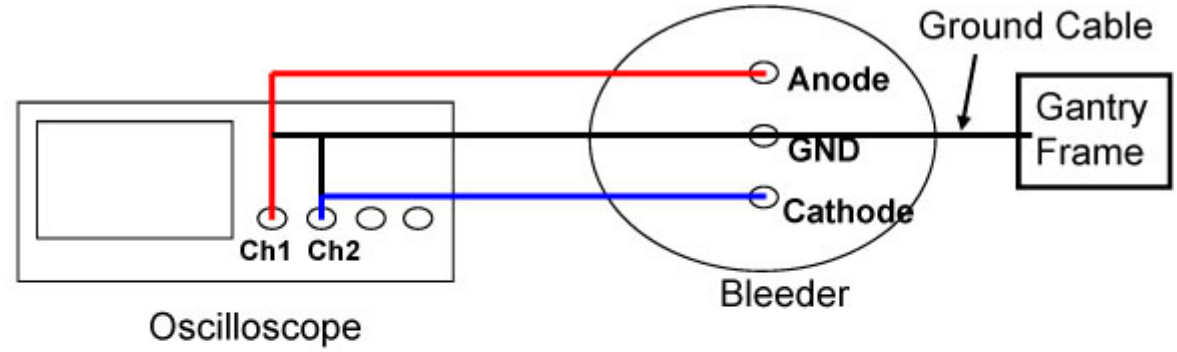
NOTICE

Potential for Equipment Damage.

Ensure that you attach a ground cable from the bleeder to the gantry GND chassis. Incorrect bleeder setup can result in damage to the HV Subsystem.

9. Unlock the gantry, and rotate the gantry such that the x-ray tube is in the 12 o'clock position.
10. Lock the gantry in position.
11. Setup the oscilloscope.
    a. To minimize noise and obtain a common ground, you should plug the oscilloscope to the bleeder.

Illustration 3: Oscilloscope Installation







b. Connect channel one probe to KV_N (TP18) and scope ground to KV_GND (TP17) on the CT Interface board in the power unit.

c. If 10x probes are used, set horizontal scale = 200 ms, vertical scale = 10V/div

d. Trigger channel one, negative slope, DC couple, trigger mode normal.

12. Turn ON the **120VAC** and **HVDC ENABLE** on the Service Switch Panel.
13. Clear the E-Stop condition by pressing the **DRIVES RESET** button on the Service Switch Panel.
14. Wait about 2 minutes for the gantry display to reset. No flashing lights on the gantry display should indicate this.
15. On the Service Desktop, select [DIAGNOSTICS], then [KV & MA (X-RAY)].
16. Verify the following scan parameters:
    - X-Ray test type = manual
    - Gantry = Disabled
    - Exposure time = 1s
    - No of scans = 1
    - Focal Spot = Large
    - V = 80kV
    - mA = 50mA
17. Press RUN and take scan.
18. Measure the output. (It may be noisy ~2v peak-to-peak. This is normal.)
    a. Using cursors on the oscilloscope, measure the mean for channel 1 from just inside the front edge to just inside the back edge of the waveform.
    b. Repeat for channel 2 mean.
19. Repeat the measurements for 100, 120, and 140kV scans.
20. Verify internal scan timer.
    a. Change trigger slope on oscilloscope to positive. Adjust trigger level to TTL if necessary.
    b. Take the following scan.
        - X-Ray test type = manual

- Gantry = Disabled
- Exposure time = 1s
- No of scans = 1
- Focal Spot = Large
- kV = 100kV
- mA = 50mA

21. Measure the output.

    Using cursors on the oscilloscope, measure the scan time (Δt) from just inside the front edge to just inside the back edge of the waveform.



NOTICE

Potential for equipment damage.

Incorrectly sealed and secured HV cables can result in damage to the tank or other parts of the gantry.

22. Reconnect HV cables from tube to tanks.

5 Finalization

1. The scale factor for your kV measurements (at the test points) is x 20 (i.e. 4V => 8kV).
2. The kV measured at the bleeder and at the test points should be within ±3% of the requested kV, respectively.
3. The internal scan timer measurement should be within ±4%.
4. Record the results on the HHS data sheet and / or take appropriate service action.

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